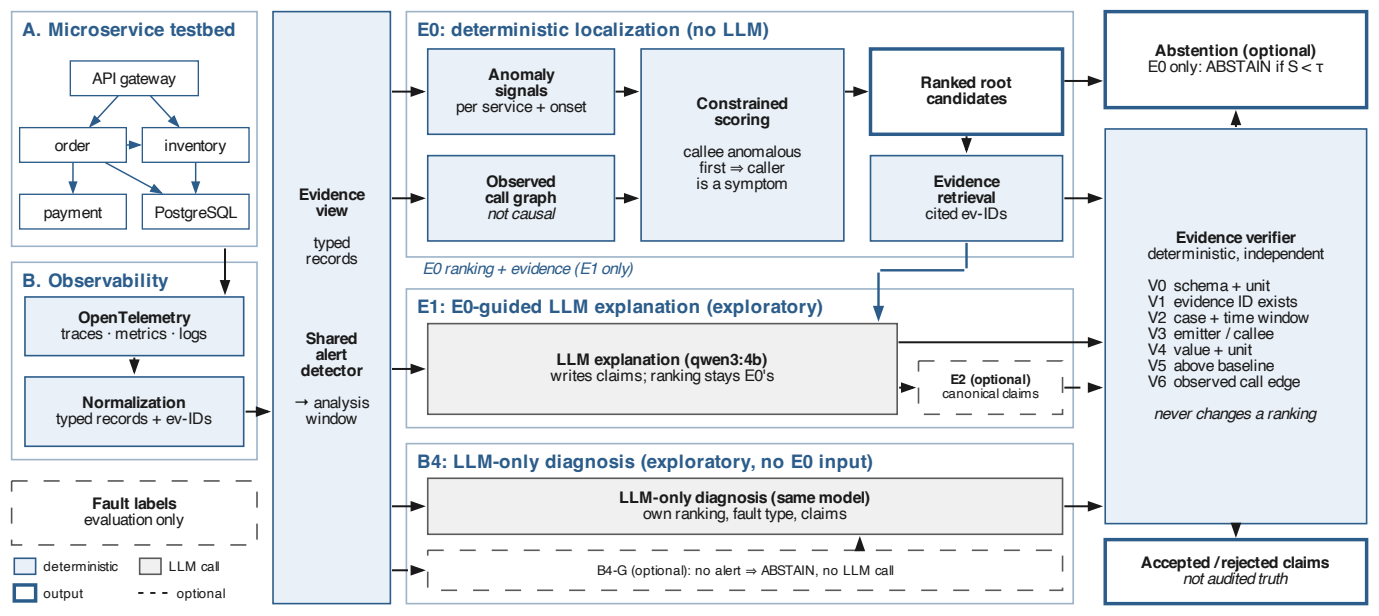

A. Microservice testbed
API gateway
order
inventory
payment
PostgreSQL
B. Observability
OpenTelemetry
traces · metrics · logs
Normalization
typed records + ev-IDs
Fault labels
evaluation only
deterministic
LLM call
output
optional
Evidence view
typed records
Shared alert detector
→ analysis window
E0: deterministic localization (no LLM)
Anomaly signals
per service + onset
Observed call graph
not causal
Constrained scoring
callee anomalous first ⇒ caller is a symptom
Ranked root candidates
Evidence retrieval
cited ev-IDs
E0 ranking + evidence (E1 only)
E1: E0-guided LLM explanation (exploratory)
LLM explanation (qwen3:4b)
writes claims; ranking stays E0's
E2 (optional)
canonical claims
B4: LLM-only diagnosis (exploratory, no E0 input)
LLM-only diagnosis (same model)
own ranking, fault type, claims
B4-G (optional): no alert ⇒ ABSTAIN, no LLM call
Abstention (optional)
E0 only: ABSTAIN if S < τ
Evidence verifier
deterministic, independent
V0 schema + unit
V1 evidence ID exists
V2 case + time window
V3 emitter / callee
V4 value + unit
V5 above baseline
V6 observed call edge
never changes a ranking
Accepted / rejected claims
not audited truth